Simon Downey

Senior Protocol Engineer

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SUMMARY

9 years designing and shipping security-critical systems: audited hardware wallet apps, consensus engines, and EVM-compatible L1s. Current work focuses on agent harnesses and autoresearch infrastructure for formally verifiable problem spaces.

Skills: C | C++ | CI/CD | Cryptography | Docker | Distributed Systems | Embedded | EVM | Git | Go | Python | Technical Leadership | TypeScript | Web3

EXPERIENCE

Core Engineering Lead, Monolythium

2025 — 2026

- Directed end-to-end design and implementation of the Monolythium blockchain, delivering a Cosmos-based chain with EVM integration and a testnet that produced over 3M blocks
  - Mono-chain - Cosmos-based chain + EVM integration
  - LythiumBFT - CometBFT w/modified selection mechanics
    - 'Quadratic Proposer Selection' - a priority-based 'concentration tax' on stake dominance

Blockchain Engineer, Solar Network

2022 — 2025

- Advised and directed protocol development
- Led technical integration across 20+ centralized exchanges, coordinating implementation requirements and production readiness
- Conducted technical interviews and mentored engineers
- Established strict operational and code-review standards
- Neutralized state-sponsored in/exfil
- Built the Solar Ledger App - Kudelski-audited
- Maintained the Python Client & Crypto SDKs

IoT Lead, ARK Ecosystem

2017 — 2022

- Developed embedded crypto architecture
- Provided technical support to users and developers
- Authored technical and community blog posts
- Represented ARK at international conferences (Geneva, NYC, San Francisco, Seoul)
- Mentored students and provided workshops at hackathons
- Maintained:
  - ARK Ledger App - cited in Ledger's security docs
  - ARK Ledger Transport
  - ARK C++ Client SDK
  - ARK C++ Crypto SDK
  - ARK Documentation

VOLUNTEER

- ARK Community Committee - *Board Member: 2017 — 2018*
- Blockhack - *Mentor: Spring 2018*
- HackPHS - *Mentor: Fall 2018*
- HackPrinceton - *Mentor: Fall 2017*, *Spring 2018*, *Fall 2018*

ARTICLES

- *'The Scalability Trilemma'*
- *'Scaling: Peeling Back the Layers'*
- *'CryptoMories - NFTs with Heart and Soul'*
- *'Expressive, Emotive NFTs - Gavin Meeler'*